Felix Zimmer

Independent Researcher

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PROFILE

Passionate about designing innovative methods. Building on a strong foundation in quantitative psychology and computational methods, I am expanding my research toward trustworthy and efficient machine learning. I thrive in interdisciplinary, collaborative environments and seek opportunities to contribute to cutting-edge advancements in ML.

WORK EXPERIENCE

Independent Researcher — 04/2023 – PRESENT

Expanding toward efficient and trustworthy AI. Developed EntryPrune – a neural network feature selection method using novel entry-based pruning that outperforms SOTA approaches. Initiated collaboration to publish a preprint and open-source package.

Freelance AI Consultant — 11/2025 – 12/2025

University of Mainz

Supporting research at the Department of Social and Legal Psychology by designing web-scraping and LLM-based content analysis tools for studies on political apologies.

Postdoctoral Researcher — 09/2023 – 12/2023 (50%)

University of Zurich

Finalized PhD-derived publications; aided departmental projects.

Visiting PhD Student — 08/2022 – 02/2023

King's College London

Collaborated with Daniel Stahl's group on sample size methods for prediction models, initiating methodology development (ongoing).

PhD Student — 04/2020 – 03/2023

University of Zurich

Developed novel sample size methods for psychological research: analytical solutions (Item Response Theory) and simulation-based approaches for complex study designs, yielding three top-tier publications.

AWARDS

- 2025 Grant to develop methods for sample size planning in prediction modeling leveraging my `mlpwr` package (Co-investigator, NIHR206858)
- 2022 Mobility grant for research stay with Prof. Daniel Stahl, King's College London (CHF 14.243, SNSF)
- 2021 Best poster award, 'MaDoKo' – Master and Doctoral Students Congress, University of Zurich
- 2020 Master thesis award for outstanding performance, University of Mainz

SOFTWARE

- entryprune — Neural network feature selection using entry-based pruning [Python, GitHub]
- mlpwr — Cost-efficient sample size planning for complex study designs [R, CRAN]
- irtpwr — Power analysis toolbox for Item Response Theory models [R, CRAN]

TEACHING

- 2022 Seminar on deep learning applications in psychology, University of Zurich [open materials]

EDUCATION

04/2020 – 04/2023 **Doctor of Philosophy** – Psychology
University of Zurich, Switzerland
Summa cum laude; Thesis: "New Methods for Power Analysis and Sample Size Planning"

04/2018 – PRESENT **Bachelor of Science** – Mathematics
FernUniversität in Hagen, Germany
Current grade: 1.8; Thesis: "How Versatile Is Hypothesis Testing for Adapting Gaussian Processes?"

10/2017 – 02/2020 **Master of Science** – Psychology
University of Mainz, Germany
Grade: 1.1; Thesis: "Applications of machine learning in psychology – a review"

10/2014 – 12/2017 **Bachelor of Science** – Psychology
University of Mainz, Germany
Grade: 1.1; Thesis: "Abstinence from Masturbation: Links to Hypersexuality, Sexual Dysfunctions, and Attitudes"

08/2005 – 03/2014 **Allgemeine Hochschulreife**
Rabanus-Maurus-Gymnasium, Mainz, Germany
Grade: 1.3

SELECTED PUBLICATIONS

Zimmer, F., Okanovic, P., & Hoefler, T. (2025). EntryPrune: Neural Network Feature Selection using First Impressions. *arXiv preprint*. [DOI]

Zimmer, F., & Debelak, R. (2023). Simulation-based design optimization for statistical power: Utilizing machine learning. *Psychological Methods*. [DOI]

Zimmer, F., Draxler, C., & Debelak, R. (2023). Power Analysis for the Wald, LR, Score, and Gradient Tests in a Marginal Maximum Likelihood Framework: Applications in IRT. *Psychometrika*. [DOI]

Zimmer, F., Henninger, M., & Debelak, R. (2023). Sample size planning for complex study designs: A tutorial for the mlpwr package. *Behavior Research Methods*. [DOI]

Zimmer, F., & Imhoff, R. (2020). Abstinence from masturbation and hypersexuality. *Archives of Sexual Behavior*. [DOI]

SELECTED TALKS

06/2023 **Conference Talk**, *Psychoco* (Zurich, Switzerland): "Minimum Sample Size Calculation for Prediction Modelling Using Surrogate Modelling"

09/2022 **Conference Talk**, *Congress of the German Psychological Society* (Hildesheim, Germany): "Simulation-based Sample Size Planning Using Machine Learning"

07/2022 **Conference Talk**, *International Meeting of the Psychometric Society* (Bologna, Italy): "Utilizing Machine Learning for Simulation-based Design Optimization"

03/2022 **Invited Colloquium Talk**, *University of Kiel* (Esther Ulitzsch), "Simulation-based design optimization for statistical power: Utilizing machine learning"

07/2021 **Conference Talk**, *European Congress of Methodology* (Valencia, Spain): "Power Analysis for the Wald, LR, Score, and Gradient Test in a Marginal Maximum Likelihood Framework: Applications in IRT"

INTERESTS

In my spare time I enjoy gravel cycling, bouldering, and eating lots of Korean ramen.